EXTENSION 2

Proof (Ext2), P1 Nature of Proof (Ext2)

Converse, Contradiction and Contrapositive Proof

Teacher: Nardene Dodd

Exam Equivalent Time: 82.5 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. Proof, EXT2 P1 2020 HSC 7 MC

Consider the proposition:

'If $2^n - 1$ is not prime, then n is not prime'.

Given that each of the following statements is true, which statement disproves the proposition?

A. $2^5 - 1$ is prime

B. $2^6 - 1$ is divisible by 9

C. $2^7 - 1$ is prime

D. $2^{11} - 1$ is divisible by 23

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2. Proof, EXT2 P1 2021 HSC 4 MC

Consider the statement:

'For all integers n, if n is a multiple of 6, then n is a multiple of 2'.

Which of the following is the contrapositive of the statement?

A. There exists an integer n such that n is a multiple of 6 and not a multiple of 2.

B. There exists an integer n such that n is a multiple of 2 and not a multiple of 6.

C. For all integers n, if n is not a multiple of 2, then n is not a multiple of 6.

D. For all integers n, if n is not a multiple of 6, then n is not a multiple of 2.

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3. Proof, EXT2 P1 2023 HSC 2 MC

Consider the following statement.

'If an animal is a herbivore, then it does not eat meat.'

Which of the following is the converse of this statement?

A. If an animal is a herbivore, then it eats meat.

B. If an animal is not a herbivore, then it eats meat.

C. If an animal eats meat, then it is not a herbivore.

D. If an animal does not eat meat, then it is a herbivore.

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4. Proof, EXT2 P1 2023 HSC 4 MC

Consider the following statement about real numbers.

'Whichever positive number r you pick, it is possible to find a number x greater than 1 such that $\dfrac{\ln x}{x^3} < r$.'

When this statement is written in the formal language of proof, which of the following is obtained?

A. $\forall x > 1 \quad \exists r > 0 \quad \dfrac{\ln x}{x^3} < r$

B. $\exists x > 1 \quad \forall r > 0 \quad \dfrac{\ln x}{x^3} < r$

C. $\forall r > 0 \quad \exists x > 1 \quad \dfrac{\ln x}{x^3} < r$

D. $\exists r > 0 \quad \forall x > 1 \quad \dfrac{\ln x}{x^3} < r$

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5. Proof, EXT2 P1 SM-Bank 1 MC

Consider the following statement.

"If you have no treasure, I have no kingdom."

Which of the following is logically equivalent to this statement?

A. If I have no kingdom then you have no treasure.

B. If you have treasure then I have a kingdom.

C. If you have no kingdom then I have no treasure.

D. If I have a kingdom then you have treasure.

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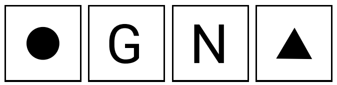
6. Proof, EXT2 P1 SM-Bank 5 MC

Four cards are placed on a table with a letter on one face and a shape on the other.



You are given the rule: "if N is on a card then a circle is on the other side."

Which cards need to be turned over to check if this rule holds?

A. N and G

B. G and triangle

C. circle and N

D. N and triangle

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7. Proof, EXT2 P1 2021 HSC 5 MC

Which of the following statements is FALSE?

A. $\forall a, b \in \mathbb{R}, \quad a < b \Rightarrow a^3 < b^3$

B. $\forall a, b \in \mathbb{R}, \quad a < b \Rightarrow e^{-a} > e^{-b}$

C. $\forall a, b \in (0, +\infty), \quad a < b \Rightarrow \ln a < \ln b$

D. $\forall a, b \in \mathbb{R}, \text{with } a, b \neq 0, \quad a < b \Rightarrow \dfrac{1}{a} > \dfrac{1}{b}$

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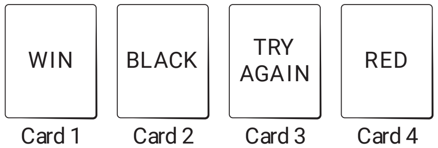
8. Proof, EXT2 P1 2021 HSC 9 MC

Four cards have either RED or BLACK on one side and either WIN or TRY AGAIN on the other side.

Sam places the four cards on the table as shown below.



A statement is made: 'If a card is RED, then it has WIN written on the other side'.

Sam wants to check if the statement is true by turning over the minimum number of cards.

Which cards should Sam turn over?

A. 1 and 4

B. 3 and 4

C. 1. 2 and 4

D. 1, 3 and 4

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9. Proof, EXT2 P1 2022 HSC 7 MC

Consider the statement P.

P: For all integers $n \geq 1$, if n is a prime number then $\dfrac{n(n+1)}{2}$ is a prime number.

Which of the following is true about this statement and its converse?

A. The statement P and its converse are both true.

B. The statement P and its converse are both false.

C. The statement P is true and its converse is false.

D. The statement P is false and its converse is true.

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10. Proof, EXT2 P1 2020 HSC 8 MC

Consider the statement:

'If n is even, then if n is a multiple of 3, then n is a multiple of 6'.

Which of the following is the negation of this statement?

A. n is odd and n is not a multiple of 3 or 6.

B. n is even and n is a multiple of 3 but not a multiple of 6.

C. If n is even, then n is not a multiple of 3 and n is not a multiple of 6.

D. If n is odd, then if n is not a multiple of 3 then n is not a multiple of 6.

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11. Proof, EXT2 P1 2021 HSC 12b

Consider Statement A.

Statement A: 'If n^2 is even, then n is even.'

i. What is the converse of Statement A?. ***(1 mark)***

ii. Show that the converse of Statement A is true. ***(1 mark)***

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12. Proof, EXT2 P1 SM-Bank 10

Prove that $\sqrt{11} - \sqrt{5} < \sqrt{2}$ by contradiction. ***(2 marks)***

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13. Proof, EXT2 P1 SM-Bank 13

If $(n-3)^2$ is an even integer, prove by contrapositive that n is odd. ***(2 marks)***

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14. Proof, EXT2 P1 SM-Bank 15

Prove $\sqrt{5} + \sqrt{3} > \sqrt{14}$ by contradiction. ***(2 marks)***

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15. Proof, EXT2 P1 SM-Bank 3

Prove that $\dfrac{1}{\sqrt{2}}$ is irrational. ***(3 marks)***

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16. Proof, EXT2 P1 2020 HSC 14d

Prove that for any integer $n > 1, \log_n(n+1)$ is irrational. ***(3 marks)***

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17. Proof, EXT2 P1 2022 HSC 13a

Prove that for all integers n with $n \geq 3$, if $2^n - 1$ is prime, then n cannot be even. ***(3 marks)***

18. Proof, EXT2 P1 2023 HSC 12a

Prove that $\sqrt{23}$ is irrational. *(3 marks)*

19. Proof, EXT2 P1 SM-Bank 11

If $a^2 - 4a + 3$ is even, $a \in \mathbb{Z}$,

prove by contrapositive that a is odd. *(3 marks)*

20. Proof, EXT2 P1 SM-Bank 12

If ab is divisible by 3, prove by contrapositive that a or b is divisible by 3. *(3 marks)*

21. Proof, EXT2 P1 SM-Bank 14

Prove that $\log_3 7$ is irrational. *(2 marks)*

22. Proof, EXT2 P1 SM-Bank 2

Prove that $\sqrt{3}$ is irrational. *(3 marks)*

23. Proof, EXT2 P1 SM-Bank 4

Prove that $\sqrt[3]{2}$ is irrational. *(3 marks)*

24. Proof, EXT2 P1 SM-Bank 8

Prove that $\log_2 11$ is irrational. *(2 marks)*

25. Proof, EXT2 P1 SM-Bank 9

If n is a positive integer,

prove $\sqrt{10n+2}$ is always irrational. *(3 marks)*

26. Proof, EXT2 P1 2020 HSC 15a

In the set of integers, let P be the proposition:

'If $k+1$ is divisible by 3, then k^3+1 divisible by 3.'

i. Prove that the proposition P is true. *(2 marks)*

ii. Write down the contrapositive of the proposition P. *(1 mark)*

iii. Write down the converse of the proposition P and state, with reasons, whether this converse is true or false. *(3 marks)*

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Worked Solutions

1. Proof, EXT2 P1 2020 HSC 7 MC

Strategy 1 – Contradiction

Consider option D,

Since $2^{11}-1$ is divisible by 23, it is NOT prime.

The proposition states that 11 is not prime which is false.

$\therefore\ 2^{11}-1$ is divisible by 23, disproves the proposition.

Strategy 2 – Contrapositive

The proposition is conditional

$X \Rightarrow Y$

Logically equivalent contrapositive statement

$\neg\, Y \Rightarrow \neg\, X$

i.e. If n is prime $\Rightarrow$ 2^n-1 is prime.

Consider D:

$n = 11$ (prime)

$2^{11}-1$ is divisible by 23 (not prime)

$\therefore$ Contrapositive statement is false and disproves the proposition.

$\Rightarrow\ D$

2. Proof, EXT2 P1 2021 HSC 4 MC

Logically equivalent statements:

If n is multiple of 6 $\Rightarrow$ n is a multiple of 2

$\neg\ n$ is a multiple of 2 $\Rightarrow$ $\neg\ n$ is a multiple of 6

i.e. if n is not multiple of 2 $\Rightarrow$ n is not a multiple of 6

$\Rightarrow\ C$

3. Proof, EXT2 P1 2023 HSC 2 MC

Statement: $P \Rightarrow Q$

Converse of statement: $Q \Rightarrow P$

$\Rightarrow D$

4. Proof, EXT2 P1 2023 HSC 4 MC

Whichever positive number r you pick ... $\forall r > 0$

It is possible to find a number x greater then 1 ... $\exists x > 1$

Such that $\dfrac{\ln x}{x^3} < r$

$\Rightarrow C$

5. Proof, EXT2 P1 SM-Bank 1 MC

The statement is conditional.
If X then Y or $X \Rightarrow Y$
The contrapositive statement is logically equivalent.
(i.e.) $\neg Y \Rightarrow \neg X$
If I have a kingdom then you have treasure.
$\Rightarrow D$

6. Proof, EXT2 P1 SM-Bank 5 MC

Solution 1
Logically equivalent statements are:

$$N \Rightarrow \text{ circle}$$
$$\neg \text{ circle} \Rightarrow \neg N$$

To confirm rule is not broken,
N must be turned
Triangle (not circle) must be turned – only other shape not a circle.
$\Rightarrow D$

Solution 2
Consider the flip side of each card.
If circle has N on the other side (or not)–tells us nothing.
If G has a circle on the other side (or not)–tells us nothing.
If N doesn't have a circle on other side–rule broken.
If triangle has an N on other side–rule broken.
$\therefore$ Need to turn N and triangle

7. Proof, EXT2 P1 2021 HSC 5 MC

By contradiction:
Consider D
Let $a = -1$ and $b = 1$,

$$a < b \ \rightarrow \ -1 < 1 \ \text{(TRUE)}$$
$$\frac{1}{a} > \frac{1}{b} \ \rightarrow \ -1 > 1 \ \text{(FALSE)}$$

$\Rightarrow$ D is false

♦ *Mean mark 50%.*

8. Proof, EXT2 P1 2021 HSC 9 MC

Logically equivalent statements are:

$$\text{Red} \Rightarrow \text{ Win} \ldots (1)$$
$$\neg \text{ Win} \Rightarrow \neg \text{ Red} \ldots (2)$$

To confirm statement is true
Card 1 – no need to turn
Card 2 – no need to turn
Card 3 – turn to confirm (2)
Card 4 – turn to confirm (1)

$\Rightarrow B$

♦ *Mean mark 48%.*

9. Proof, EXT2 P1 2022 HSC 7 MC

Statement: $\forall n \in \mathbb{Z}^+$, if n is prime $\Rightarrow \dfrac{n(n+1)}{2}$ is prime.

Converse: $\forall n \in \mathbb{Z}^+$, if $\dfrac{n(n+1)}{2}$ is prime $\Rightarrow$ n is prime.

Consider prime $n = 3$:

$$\frac{n(n+1)}{2} = \frac{3 \times 4}{2} = 6 \ \text{(not prime} \rightarrow \text{Statement is false)}$$

Consider $\dfrac{n(n+1)}{2}$:

$$\frac{n(n+1)}{2} \text{ is a prime} \Leftrightarrow n = 2 \ \text{(prime)}$$

$\therefore$ Converse is true
$\Rightarrow D$

♦♦ *Mean mark 37%.*

10. Proof, EXT2 P1 2020 HSC 8 MC

Proposition: If $X \Rightarrow Y$
X is a compound statement
"If n is even and a multiple of 3."
Y states "n is a multiple of 6."

Negation if X but $\neg Y$.

$\Rightarrow B$

♦♦ *Mean mark part 33%.*

11. Proof, EXT2 P1 2021 HSC 12b

i. Converse

If n is even, then n^2 is even.

ii. If n is even:

$$n = 2p,\ p \in \mathbb{Z}$$

$$\begin{aligned} n^2 &= (2p)^2 \\ &= 4p^2 \\ &= 2(2p^2) \\ &= 2q,\ q \in \mathbb{Z} \end{aligned}$$

$\therefore$ If n is even, then n^2 is even.

12. Proof, EXT2 P1 SM-Bank 10

Proof by contradiction:

Assume that $\sqrt{11} - \sqrt{5} \geq \sqrt{2}$

$$\left(\sqrt{11} - \sqrt{5}\right)^2 \geq 2$$

$$11 - 2\sqrt{55} + 5 \geq 2$$

$$2\sqrt{55} \leq 14$$

$$\sqrt{55} \leq 7$$

$$55 \leq 49 \quad \text{(which is incorrect)}$$

$\therefore$ By contradiction, $\sqrt{11} - \sqrt{5} < \sqrt{2}$

13. Proof, EXT2 P1 SM-Bank 13

Statement

If $(n-3)^2$ is even $\Rightarrow n$ is odd

Contrapositive

If $n \neg$ odd $\Rightarrow (n-3)^2 \neg$ even

(i.e.) n even $\Rightarrow (n-3)^2$ is odd

If n even, $\exists\, k,\ k \in \mathrm{Z}$ where $n = 2k$

$$\begin{aligned} (n-3)^2 &= (2k-3)^2 \\ &= 4k^2 - 12k + 9 \\ &= 4\left(k^2 - 12k + 2\right) + 1 \end{aligned}$$

$\Rightarrow (n-3)^2$ is odd

$\therefore$ If n is even, $(n-3)^2$ is odd

$\therefore$ By contrapositive, statement is true.

14. Proof, EXT2 P1 SM-Bank 15

Proof by contradiction:

Assume $\sqrt{5} + \sqrt{3} \leq \sqrt{14}$

$$\left(\sqrt{5} + \sqrt{3}\right)^2 \leq \left(\sqrt{14}\right)^2$$

$$5 + 2\sqrt{15} + 3 \leq 14$$

$$2\sqrt{15} \leq 6$$

$$\sqrt{15} \leq 3$$

$$15 \leq 9 \quad \text{(incorrect)}$$

$\therefore$ By contradiction, $\sqrt{5} + \sqrt{3} > \sqrt{14}$

15. Proof, EXT2 P1 SM-Bank 3

Proof by contradiction:

Assume that $\dfrac{1}{\sqrt{2}}$ is rational.

$\therefore \dfrac{1}{\sqrt{2}} = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$\frac{1}{2} = \frac{p^2}{q^2}$$

$$q^2 = 2p^2 \ldots (1)$$

q^2 is even $\Rightarrow$ q is even

(i.e.) $\exists k,\ k \in \mathbb{Z}$ such that $q = 2k$

Substitute $q = 2k$ into (1)

$$(2k)^2 = 2p^2$$

$$2k^2 = p^2$$

p^2 is even $\Rightarrow$ p is even

$\therefore p$ and q have a common factor of 2

$\Rightarrow$ Contradiction

$\therefore \dfrac{1}{\sqrt{2}}$ is irrational.

16. Proof, EXT2 P1 2020 HSC 14d

Proof by contradiction:

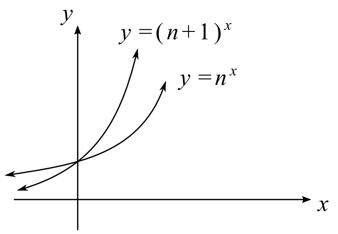
Assume $\log_n(n+1)$ is rational

$\therefore \log_n(n+1) = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$n^{\frac{p}{q}} = n+1$$

$$n^p = (n+1)^q$$

Strategy 1



$n^p = (n+1)^q$ when $p = q = 0$ only

$q \neq 0$

$\therefore$ By contradiction, $\log_n(n+1)$ is irrational.

Strategy 2

$n^p = (n+1)^q$

If n is odd, LHS is odd and RHS is even.

If n is even LHS is even and RHS is odd.

Statement is true for $p = q = 0$, but $q \neq 0$

$\therefore$ By contradiction, $\log_n(n+1)$ is irrational.

17. Proof, EXT2 P1 2022 HSC 13a

Contrapositive statement:

If n is even, $2^n - 1$ is NOT prime.

Let $n = 2k$, $(k \in \mathbb{Z}$ and $k \geq 2)$

$$\begin{aligned} 2^n - 1 &= 2^{2k} - 1 \\ &= \left(2^k\right)^2 - 1 \\ &= \left(2^k - 1\right)\left(2^k + 1\right) \end{aligned}$$

Since $k \geq 2$ $\Rightarrow$ $2^k - 1 \geq 3$ and $2^k + 1 \geq 5$

$\therefore 2^n - 1$ is not prime if n is even, as it has two non-trivial integer factors.

$\therefore$ By contrapositive statement, if $2^n - 1$ is prime, n cannot be even.

18. Proof, EXT2 P1 2023 HSC 12a

Proof by contradiction:

Assume $\sqrt{23}$ is rational

$\sqrt{23} = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$23 = \frac{p^2}{q^2}$$

$$23q^2 = p^2$$

$\Rightarrow$ 23 is a factor of p^2
$\Rightarrow$ 23 is a factor of p

$\exists k \in \mathbb{Z}$ such that $p = 23k$

$$23q^2 = (23k)^2$$
$$q^2 = 23k^2$$

$\Rightarrow$ 23 is a factor of q^2
$\Rightarrow$ 23 is a factor of q
$\therefore$ HCF ≥ 23
$\therefore$ By contradiction, $\sqrt{23}$ is rational

19. Proof, EXT2 P1 SM-Bank 11

Proof by contrapositive

$$a \neg \text{ odd } \Rightarrow a^2 - 4x + 3 \neg \text{ even}$$

(i.e) If a is even $\Rightarrow a^2 - 4x + 3$ is odd

If a is even, $\exists\ k, k \in \mathrm{Z}$, such that $a = 2k$
Substitute $2k$ into $a^2 - 4x + 3$

$$\begin{aligned}(2k)^2 - 4(2k) + 3 &= 4k^2 - 8k + 3\\ &= 2\left(2k^2 - 4k + 1\right) + 1\end{aligned}$$

$\therefore$ By contrapositive, if $a^2 - 4x + 3$ is even $\Rightarrow$ a is odd.

20. Proof, EXT2 P1 SM-Bank 12

Statement
ab is divisible by 3 $\Rightarrow a$ or b is divisible by 3
Contrapositive
a or $b \neg$ divisible by 3 $\Rightarrow ab \neg$ divisible by 3

Let $a = 3x + p$, where $x \in \mathrm{Z}$ and $p = 1$ or 2
$b = 3y + q$, where $y \in \mathrm{Z}$ and $q = 1$ or 2

$$\begin{aligned}ab &= (3x+p)(3y+q)\\ &= 9xy + 3qx + 3py + pq\\ &= 3(3xy + qx + py) + pq\end{aligned}$$

Possible values of $pq = 1, 2, 4$
$\Rightarrow$ pq is not divisible by 3
$\therefore$ ab is not divisible by 3
$\therefore$ By contrapositive, statement is true.

21. Proof, EXT2 P1 SM-Bank 14

Proof by contradiction:
Assume that $\log_3 7$ is rational.
i.e. $\log_3 7 = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1.

$$\log_3 7 = \frac{p}{q}$$
$$q\log_3 7 = p$$
$$\log_3 7^q = p$$
$$7^q = 3^p$$

Since 7 and 3 are prime numbers, $7^q \neq 3^p$
$\therefore$ Contradiction: integer values p, q do not exist.
$\therefore$ $\log_3 7$ is irrational.

22. Proof, EXT2 P1 SM-Bank 2

Proof by contradiction:

Assume that $\sqrt{3}$ is rational.

$\therefore \sqrt{3} = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$3 = \frac{p^2}{q^2}$$

$3q^2 = p^2 \ldots (1)$

If q^2 is even:

$\Rightarrow q$ is even $(2/q)$

$\Rightarrow p^2$ is even $\Rightarrow p$ is even $(2/p)$

p, q have a common factor of 2

Contradiction $\Rightarrow \sqrt{3}$ is irrational for p, q even.

COMMENT: $(2/q)$ can be used for "2 divides q" or q is divisible by 2.

If q^2 is odd:

$\Rightarrow 3q^2$ is odd $\Rightarrow q$ is odd

$\Rightarrow p^2$ is odd $\Rightarrow p$ is odd

Let $p = 2x+1, \quad q = 2y+1, \quad x, y \in \mathbb{Z}$

Substitute into (1)

$$3(2y+1)^2 = (2x+1)^2$$
$$3\left(4y^2+4y+1\right) = 4x^2+4x+1$$
$$12y^2+12y+3 = 4x^2+4x$$
$$6y^2+6y+1 = 2x^2+2x$$

$\text{LHS} = 6\left(y^2+y\right)+1$ which is odd

$\text{RHS} = 2\left(x^2+x\right)$ which is even

Contradiction $\Rightarrow \sqrt{3}$ is irrational for p, q odd.

$\therefore \sqrt{3}$ is irrational.

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23. Proof, EXT2 P1 SM-Bank 4

Proof by contradiction:

Assume that $\sqrt[3]{2}$ is rational.

$\therefore \sqrt[3]{2} = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$2 = \frac{p^3}{q^3}$$

$2q^3 = p^3 \ldots (1)$

p^3 is even $\Rightarrow p$ is even

(i.e.) $\exists k, \ k \in \mathbb{Z}$ such that $p = 2k$

Substitute $q = 2k$ into (1)

$$2q^3 = (2k)^3$$
$$2q^3 = 8k^3$$
$$q^3 = 4k^3$$

q^3 is even $\Rightarrow q$ is even

$\therefore p$ and q have a common factor of 2

$\Rightarrow$ Contradiction

$\therefore \sqrt[3]{2}$ is irrational.

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24. Proof, EXT2 P1 SM-Bank 8

Proof by contradiction:

Assume that $\log_2 11$ is rational.

$\therefore \log_2 11 = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$q\log_2 11 = p$$
$$\log_2 11^q = p$$
$$11^q = 2^p$$

LHS is odd.

RHS is even.

$\therefore$ Contradiction: integer values p, q do not exist.

$\therefore \log_2 11$ is irrational.

25. Proof, EXT2 P1 SM-Bank 9

Proof by contradiction:

Assume that $\sqrt{10n+2}$ is rational.

$\therefore \sqrt{10n+2} = \dfrac{p}{q}$ where $p, q \in \mathbb{Z}$ with no common factor except 1

$$10n+2 = \frac{p^2}{q^2}$$

$2q^2(5n+1) = p^2 \ldots (1)$

p^2 is even $\Rightarrow$ p is even

(i.e.) $\exists k,\ \ k \in \mathbb{Z}$ such that $p = 2k$

Substitute $p = 2k$ into (1)

$$2q^2(5n+1) = (2k)^2$$

$$q^2(5n+1) = 2k^2$$

$q^2(5n+1)$ is even since $\Rightarrow$ $2k^2$ is even

$\Rightarrow q^2$ is even since $5n+1$ can be odd or even.

$\Rightarrow q$ is even

$\Rightarrow$ Contradiction: p and q have a common factor of 2

$\therefore \sqrt{10n+2}$ is irrational.

26. Proof, EXT2 P1 2020 HSC 15a

i. Let $k+1 = 3N,\ N \in Z$

$\Rightarrow k = 3N – 1$

$$\begin{aligned} k^3+1 &= (3N-1)^3+1 \\ &= (3N)^3 + 3(3N)^2(-1) + 3(3N)(-1)^2 + (-1)^3 + 1 \\ &= 27N^3 – 27N^2 + 9N – 1 + 1 \\ &= 3(9N^3 – 9N^2 + 3N) \\ &= 3Q\ ,\ Q \in Z \end{aligned}$$

$\therefore$ If $k+1$ is divisible by 3, then k^3+1 is divisible by 3.

ii. Contrapositive

If k^3+1 is not divisible by 3, then $k+1$ is not divisible by 3.

iii. Converse:

♦♦ *Mean mark part (iii) 36%.*

If k^3+1 is divisible by 3, then $k+1$ is divisible by 3.

Contrapositive of converse:

If $k+1$ is not divisible by 3, then k^3+1 is not divisible by 3.

i.e. $k+1$ is not divisible by 3 when $k+1 = 3Q+1$ or $k+1 = 3Q+2$, where $Q \in Z$

If $k+1 = 3Q+1 \ \Rightarrow\ k = 3Q$

$$\begin{aligned} k^3+1 &= (3Q)^3+1 \\ &= 27Q^3+1 \\ &= 3(9Q^3)+1 \\ &= 3M+1 \ \ \text{(not divisible by 3, } M \in Z) \end{aligned}$$

If $k+1 = 3Q+2 \ \Rightarrow\ k = 3Q+1$

$$\begin{aligned} k^3+1 &= (3Q+1)^3+1 \\ &= (3Q)^3 + 3(3Q)^2 + 3(3Q) + 1 + 1 \\ &= 27Q^3 + 27Q^2 + 9Q + 2 \\ &= 3(9Q^3 + 9Q^2 + 3Q) + 2 \\ &= 3M+2 \ \text{(not divisible by 3, } M \in Z) \end{aligned}$$

$\therefore$ By contrapositive, if k^3+1 is divisible by 3, k + 1 is divisible by 3.

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